Anime Trends & Type Prediction Using Text Mining

AIM (Introduction)

This study aims to analyze anime genre patterns using text mining techniques and develop machine learning models to predict anime type such as TV, Movie, OVA, ONA, and Special. The project focuses on extracting meaningful insights from textual genre data to better understand audience preferences.

Text Mining Methods Used: Sentiment Analysis (VADER)

Purpose: To measure emotional tone of anime synopses

Why chosen: Fast, lexicon-based, works well on short/medium text, No training required.

Machine Learning Models

- Task: Predict anime type (TV, Movie, OVA, etc.)

Models tested: Logistic Regression (Best); Support Vector Machine (SVM), Random Forest

- Best Model:

Logistic Regression: achieved the highest overall performance, with strong accuracy and balanced precision and recall across anime types

Dataset Description & Use

We used the Tidy Anime (TidyTuesday 2019-04-23) dataset, which contains structured metadata about anime titles including:

- Synopsis (text descriptions)

Genre and type (TV, Movie, OVA, etc.)

Ratings and popularity metrics

- Why this dataset?

Contains rich natural language synopses

Over 1,000+ entries (meets assignment requirement)

Suitable for classification + topic modeling

Text Preprocessing & Feature Engineering

Preprocessing steps:

Lowercasing text

Removing punctuation and numbers

Removing extra whitespace

Stopword filtering (TF-IDF & CountVectorizer)

Feature Engineering:

TF-IDF vectorization (for classification)

Count Vectorization (for topic modeling)

Sentiment scoring using VADER

Topic Modeling (LDA)

- Purpose: To discover hidden themes in anime descriptions
- Why chosen: Unsupervised learning (no labels needed), Interpretable topic clusters, Ideal for exploratory text analysis

Key Results & Visualizations:

- Sentiment Distribution

Most anime synopses are neutral to slightly positive

- Topic Modeling Insights:

Example topics discovered:

Action / battle / powers / hero arcs

School / romance / relationships

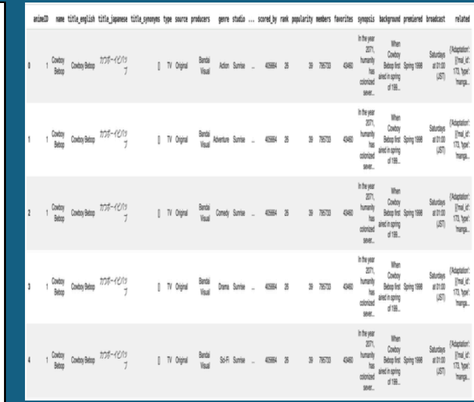
Fantasy / magic / worlds / adventure

Sci-fi / technology / future themes

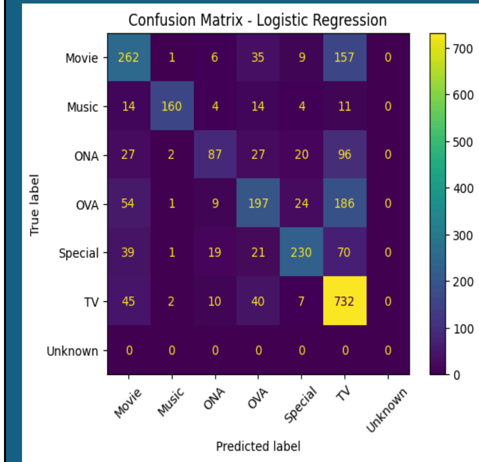
- Model Performance:

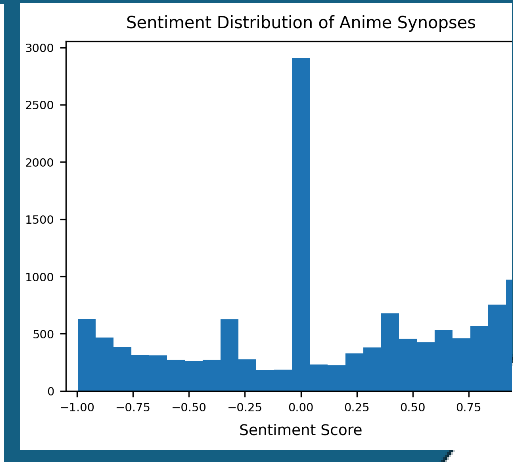
Logistic Regression achieved the best overall performance, with an accuracy of approximately 64% and balanced precision and recall across most anime types.

Performance varied across classes.



	precision	recall	f1-score	support
Movie	0.59	0.56	0.58	470
Music	0.96	0.77	0.86	207
ONA	0.64	0.34	0.44	259
OVA	0.59	0.42	0.49	471
Special	0.78	0.61	0.68	380
TV	0.58	0.88	0.70	836
accuracy			0.64	2623
macro avg	0.69	0.59	0.62	2623
weighted avg	0.65	0.64	0.62	2623





Main Insights & Conclusions

Anime synopses strongly cluster into clear thematic groups

NLP methods successfully extract meaningful structure from text

Sentiment is mostly neutral-positive → storytelling driven content

Logistic Regression is highly effective for text classification

GitHub:

Dataset:

References:

- TidyTuesday Project Dataset: Tidy Anime (TidyTuesday 2019-04-23)
- Scikit-learn Documentation (2024)
- Bird, Klein & Loper (2009) – NLTK Book
- Blei et al. (2003) – Latent Dirichlet Allocation
- Hutto & Gilbert (2014) – VADER Sentiment Analysis